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Department of Computer Science & Engineering

Sales Performance Analysis and Prediction for Zudio Retail Stores

By

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2023800127

Project

Python Programming for Data Science

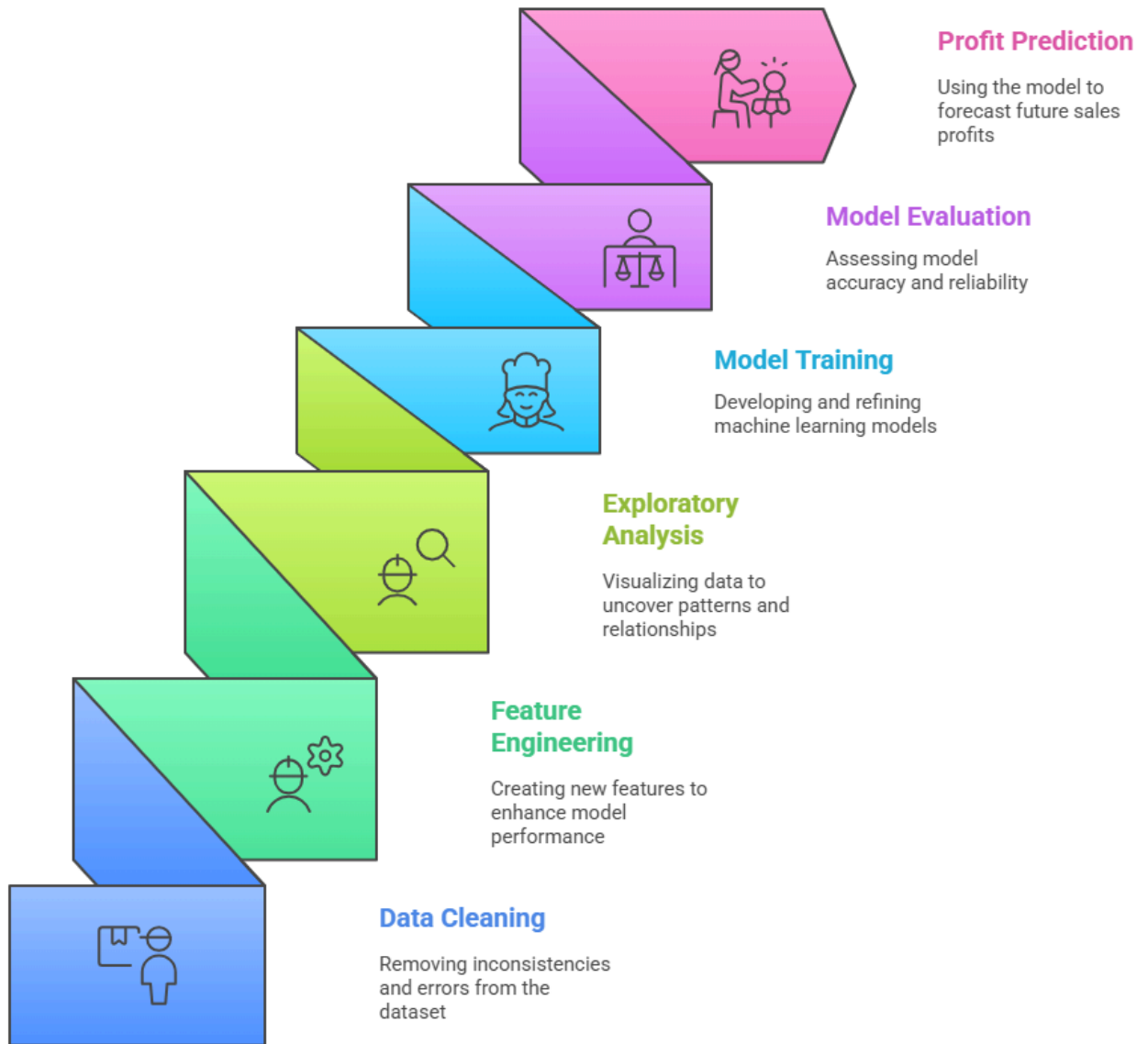
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Abstract

- Focus: Analyze and predict sales performance for Zudio retail stores using data analytics and machine learning.
- Objective: Identify sales trends, uncover profit-driving factors, and predict future sales profits using features like product type, city, category, quantity sold, and order date.
- Methodology:
 - Data preprocessing: Handled missing values, removed outliers using IQR method.
 - Feature Engineering: Extracted features like order weekdays, store age.
 - Exploratory Data Analysis (EDA): Used univariate, bivariate, and multivariate visualizations.
 - Models: Trained Linear Regression, Random Forest, XGBoost, Light GBM, and CatBoost.
 - Evaluation: Used R² scores and 5-fold cross-validation.
- Results: CatBoost achieved the highest average cross-validation R² score of 0.8742.
- Application: Predicted profit for a hypothetical scenario, aiding inventory planning, store evaluation, and demand forecasting.

Achieving Accurate Sales Prediction



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Introduction

Problem Statement:

In today's competitive retail landscape, understanding consumer buying behavior and store performance is vital for maximizing profitability. Zudio, a prominent fashion retail brand, deals with massive sales data across multiple cities and product categories. However, raw transactional data lacks actionable insights. The core problem is to analyze this sales data to discover patterns and build a model that can accurately predict sales profit based on various store and order attributes.

Objective:

- Perform EDA to identify sales trends.
- Develop a machine learning model to predict sales profit.
- Select the best model among Linear Regression, Random Forest, XGBoost, LightGBM, and CatBoost based on R2 scores.

Motivation:

Predictive analytics in the retail sector plays a crucial role in optimizing operations, improving inventory decisions, and enhancing customer satisfaction. By predicting sales profit accurately, Zudio can make data-driven decisions regarding product stocking, pricing, and marketing strategies—directly contributing to business growth. This project is also a hands-on opportunity to apply data science techniques to real-world retail data, making it both educational and impactful.

Outline:

- Data preprocessing and cleaning.
- Exploratory data analysis for trend identification.
- Feature engineering and outlier treatment.
- Machine learning model implementation and comparison.
- Profit prediction using the best model.

Dataset

Description of Dataset:

The dataset used in this project is a retail sales dataset provided by Kaggle. The Zudio-dataset consists of transaction-level data gathered from multiple Zudio retail stores across various cities in India. The dataset includes 7899 rows and 28 columns (before preprocessing), capturing key attributes such as Quantity, Sales Profit, Order Date, Store Open Date, City, Clothing Type, Category, and other order-specific details. The data is structured in tabular format, with a mix of categorical, numerical, and date-time features.

Preprocessing:

To prepare the data for analysis and modeling, the following preprocessing steps were performed:

- String "None" values were replaced with NaN
- Missing values in categorical columns were imputed using the mode (most frequent value).
- Rows with missing or corrupt date entries were dropped or converted using `pd.to_datetime` with error handling.
- Duplicate Removal:
 - Aim was to remove duplicate rows from the dataset. But in my dataset the count of duplicate rows was 0.

- Outlier Treatment:

Interquartile Range (IQR) method was applied to Quantity and Sales Profit to detect and remove outliers.

In the “Quantity” column, no outliers were detected.
But in the “Sales Profit” column, outliers were detected.

Outliers detected: 29

Value range for outliers: < -3327.60 or > 9309.20

Min outlier value: 9312.0

Max outlier value: 9596.8

- Feature Engineering:

New features were derived from existing date columns:

'Order Year', 'Order Month', 'Order Day', 'Order Weekday', 'Store Age (Days)',
'Profit per Item' (SalesProfit/Quantity, handling zero division)

Advanced feature engineering:

1. Label Count Encoding for 'City' => A technique where categorical values are replaced by their frequency (count) of occurrence in the dataset.
2. Statistical Polynomial Aggregate (SPA) for 'Sales_Quantity_SPA' (Sales * Quantity²)
=> SPA involves aggregating numerical features using polynomial-based statistical functions (mean, variance, skewness, kurtosis, etc.) grouped by categorical features.

- Encoding and Standardization:

- One-Hot Encoding: Convert 'Clothing Type' and 'Category' into binary columns for each unique value to enable algorithms to process categorical data numerically.
- Target Encoding: Replace each 'City' with the average target (sales) for that city to reduce the number of columns and retain target-related information.
- Standardization: Scale numerical features to have zero mean and unit variance using StandardScaler to improve model convergence and performance.

Data Exploration:

A comprehensive Exploratory Data Analysis (EDA) was conducted on the Zudio retail sales dataset to understand the structure, patterns, and potential insights within the data. Below are the steps and findings:

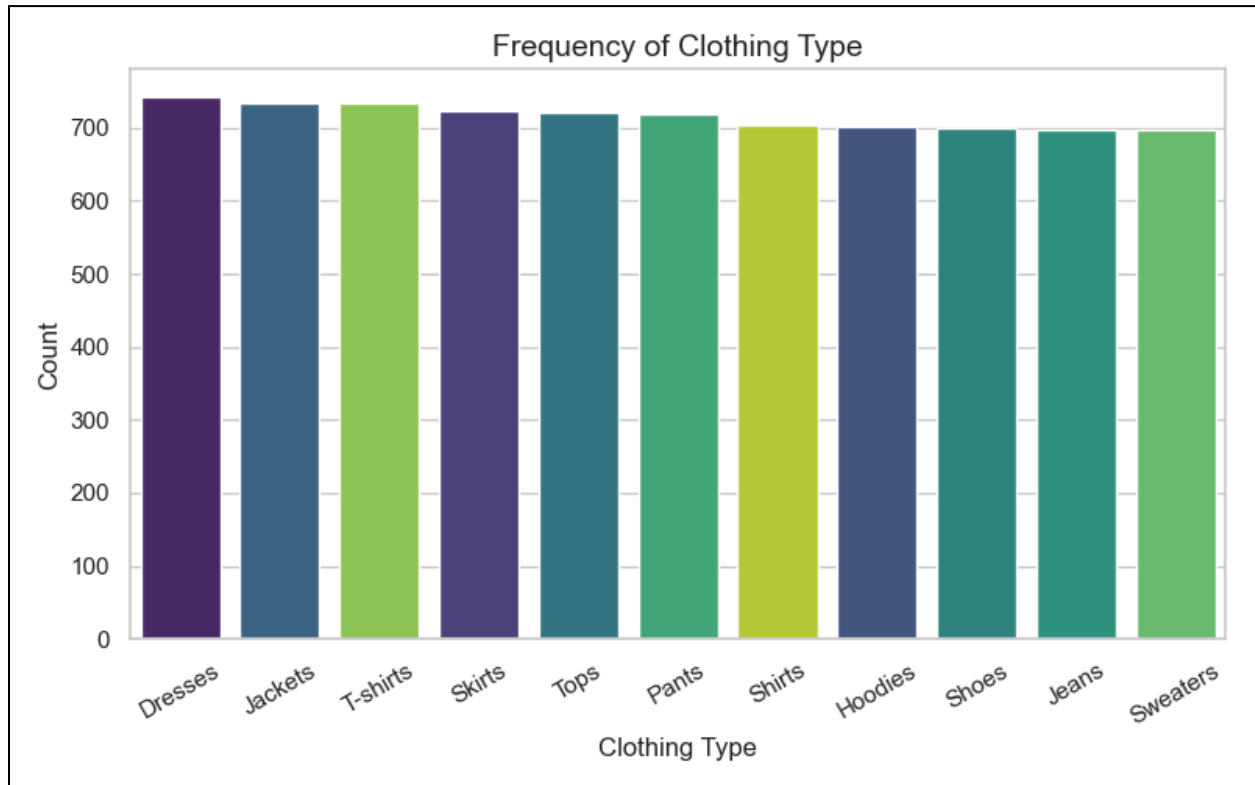
1. Summary Statistics & Structure

- Shape of Data: The dataset contains rows = `df.shape[0]` and columns = `df.shape[1]` ie 7899 rows and 28 columns
- Descriptive Statistics: `.describe()` was used to view measures like mean, median, min, max, and standard deviation for numerical features.

2. Univariate analysis

- Univariate analysis looks at one variable at a time to understand its distribution (e.g., frequency of categories, histograms).

Figure 1



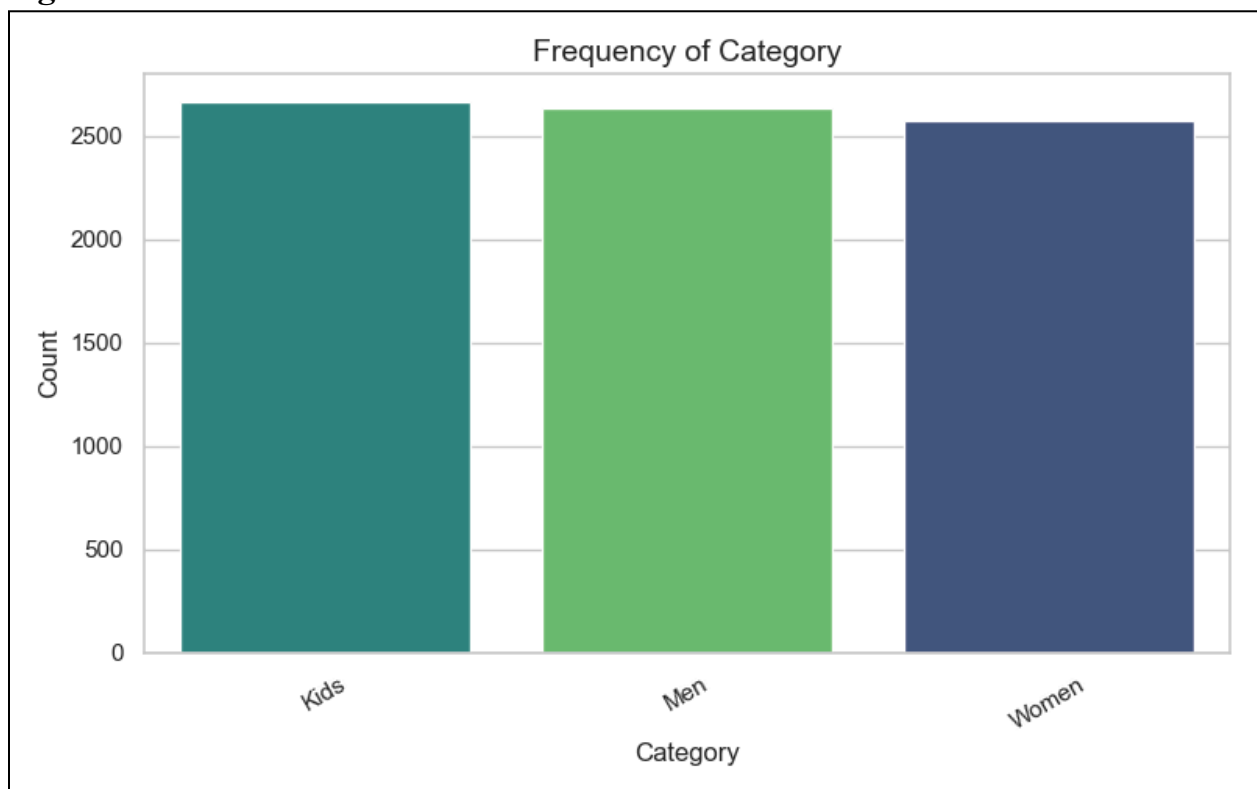
Analysis:

- This bar chart illustrates the frequency of sales across different clothing types: Dresses, Jackets, T-shirts, Skirts, Tops, Pants, Shirts, Hoodies, Shoes, Jeans, and Sweaters.
- All clothing types have similar frequencies, ranging between 700 and 750, with Dresses slightly leading the count.
- The balanced frequencies suggest that Zudio maintains a diversified inventory or that customer preferences are evenly distributed across clothing types.
- This uniformity could reflect a strategic merchandising decision to cater to a wide range of customer needs, ensuring no single clothing type dominates sales.

Justification for Plot Choice:

- A bar chart is ideal for this univariate analysis because it clearly shows the comparison among discrete clothing types. It allows easy identification of which clothing types are most and least frequent, which is crucial for inventory planning and customer preference analysis.

Figure 2



Analysis:

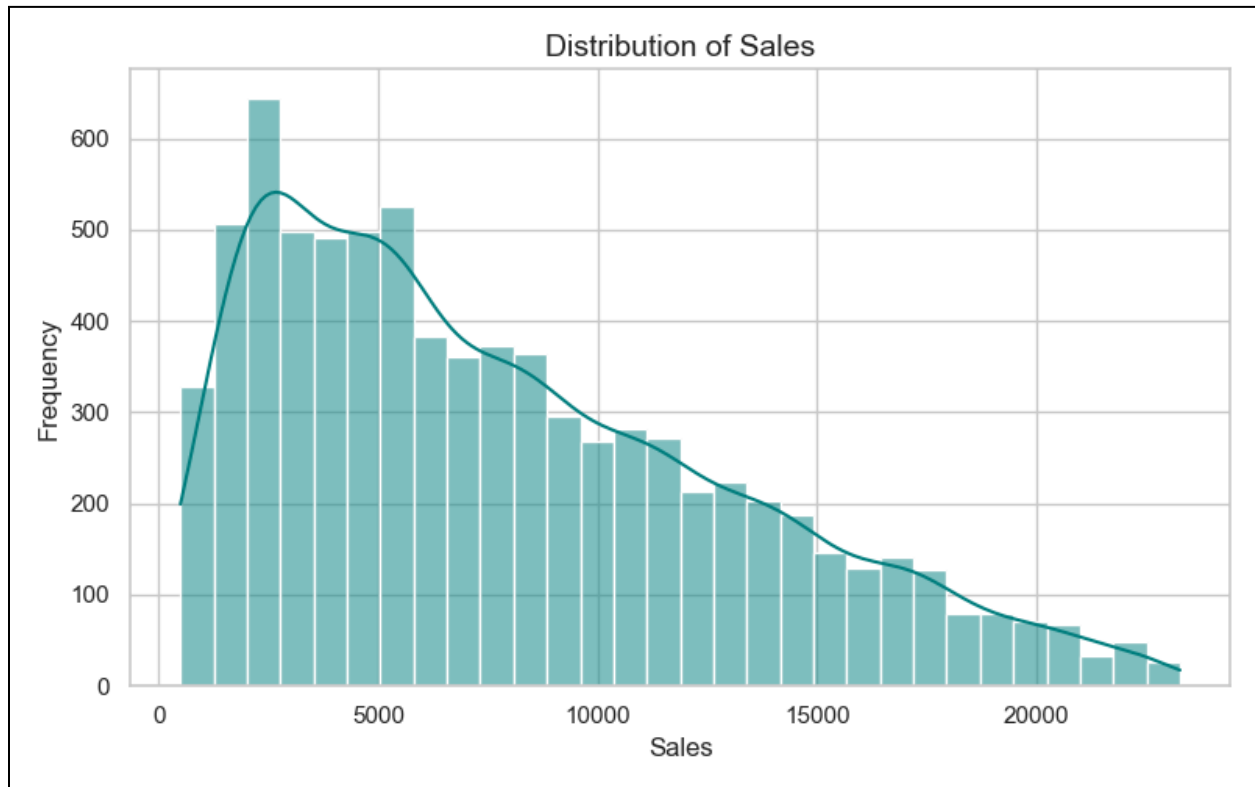
- This bar chart displays the frequency of sales across three categories: Kids, Men, and Women.
- Kids have the highest frequency, with a count slightly above 2500, indicating the highest demand for kids' apparel.

- Men and Women categories follow closely, each with counts just below 2500, showing a relatively balanced distribution but slightly less demand compared to Kids.
- The similar frequencies across categories suggest that Zudio's customer base has diverse preferences, with a slight inclination toward kids' clothing, possibly reflecting a family-oriented target market.

Justification for Plot Choice:

- A bar chart is appropriate for this univariate analysis because it effectively compares the frequency of a single categorical variable (Category). It provides a clear visual representation of the distribution, making it easy to identify which category is most frequent, which is essential for inventory planning and understanding customer preferences.

Figure 3



Analysis:

- This histogram with a kernel density estimate (KDE) shows the distribution of the Sales variable.
- There are fewer high-value sales, with the frequency dropping significantly after 10,000, and very few sales exceeding 15,000.
- The KDE curve confirms the right skew, with a long tail extending toward higher sales values, suggesting occasional large transactions but a predominance of smaller ones.
- This distribution implies that Zudio's sales are mostly driven by smaller purchases, which could influence pricing strategies or promotional offers targeting lower-value transactions.

Justification for Plot Choice:

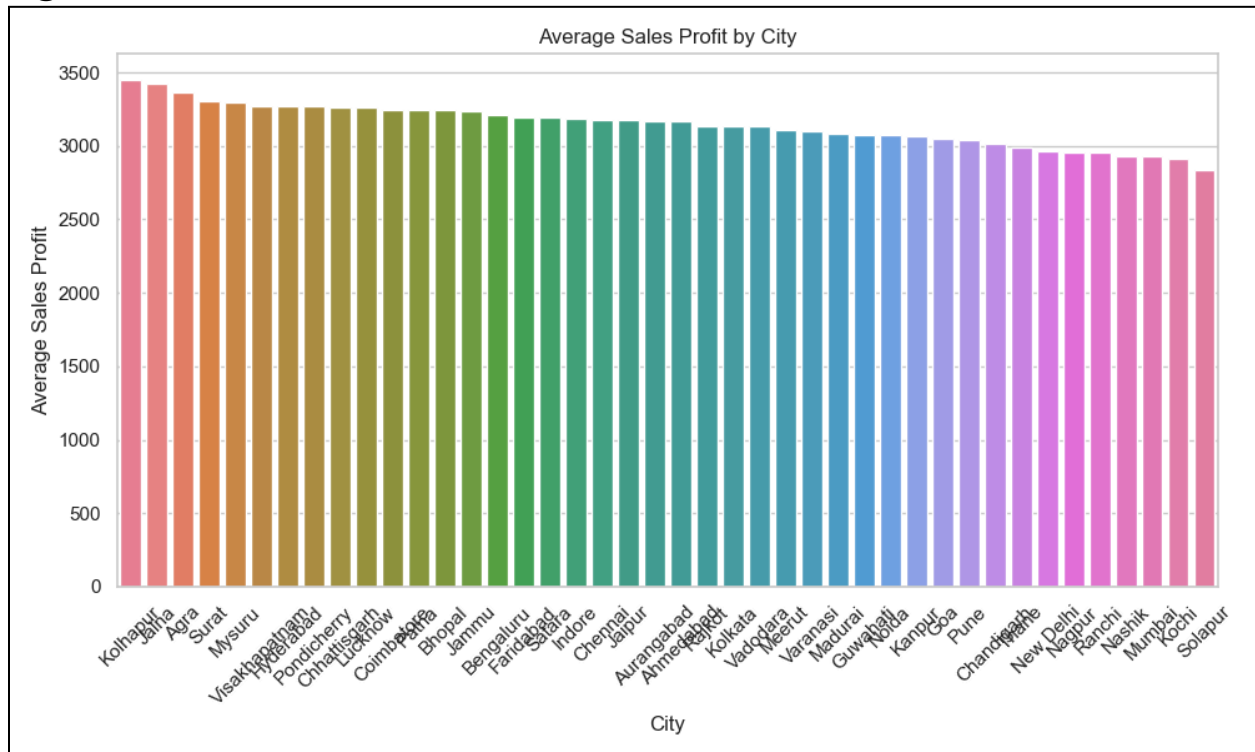
- A histogram with a KDE is a suitable choice for this univariate analysis because it provides a detailed view of the distribution of a single numerical variable (Sales). The histogram shows the frequency of sales values, while

the KDE smooths the distribution, making it easier to identify the overall shape (e.g., skewness), which is critical for understanding sales patterns and informing business decisions.

3. Bivariate analysis

- Bivariate analysis explores the relationship between two variables, like correlation or scatter plots.

Figure 1



Analysis:

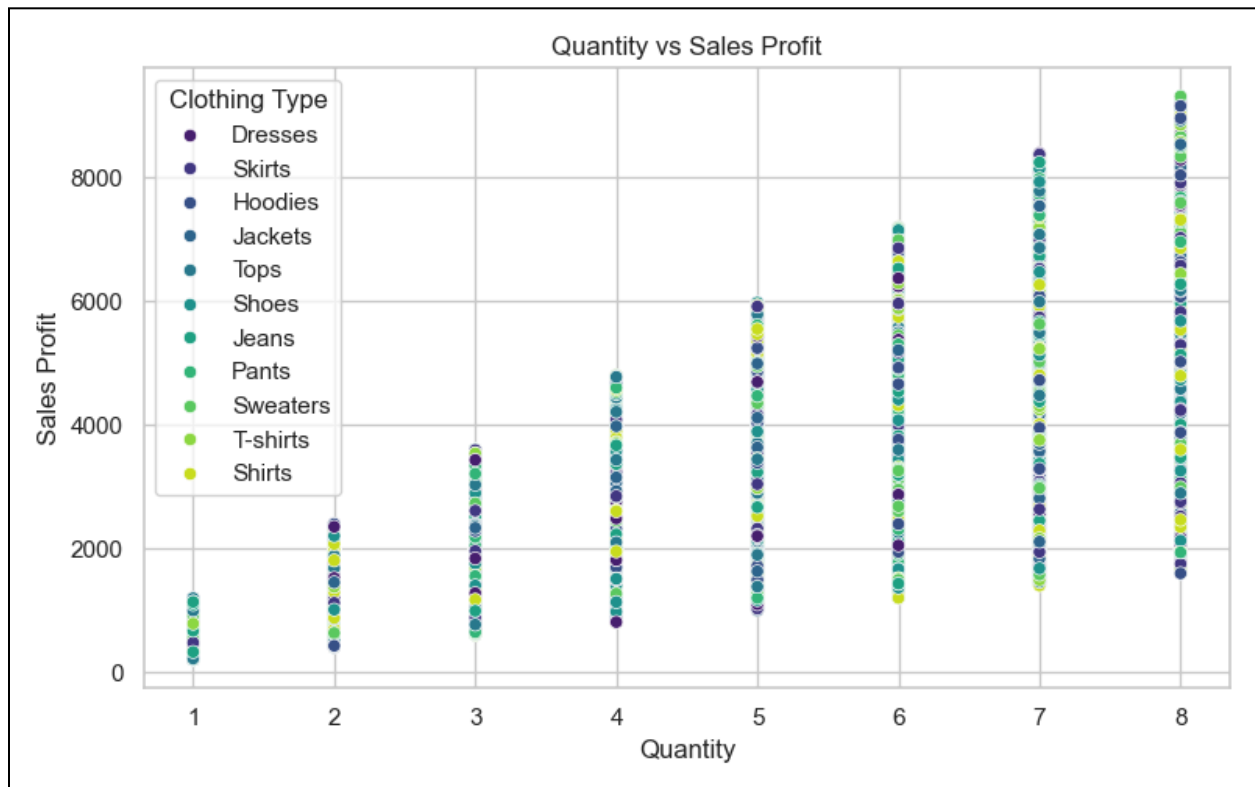
- This bar chart displays the average sales profit across different cities.
- Kolhapur, Jalna, and Agra lead with the highest average profits, around 3400, indicating strong performance in these regions.
- Solapur and Kochi have the lowest average profits, around 3000, suggesting weaker sales performance in these cities.

- Most cities show profits between 3100 and 3300, indicating a relatively consistent performance across regions, with some outliers.
- This variation highlights regional differences in sales performance, which could be due to factors like customer demographics, store location, or local economic conditions.

Justification for Plot Choice:

- A bar chart is appropriate for this bivariate analysis because it compares a numerical variable (Average Sales Profit) across a categorical variable (City). It allows for easy identification of high- and low-performing cities, which is useful for regional performance evaluation and targeted business strategies.

Figure 2



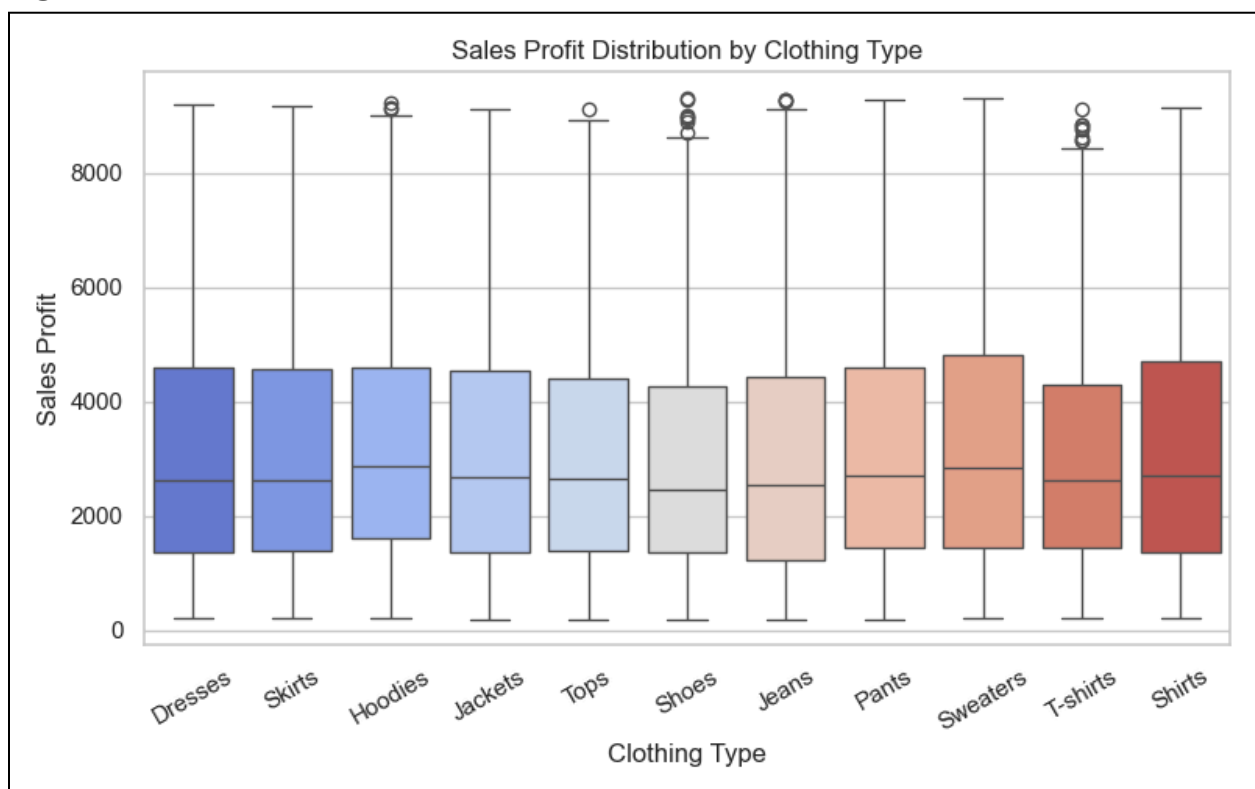
Analysis:

- This scatter plot examines the relationship between Quantity and Sales Profit, with points colored by Clothing Type.
- A general positive trend is observed: as quantity increases, sales profit tends to increase, though the relationship is not perfectly linear.
- Higher quantities (6-8) are associated with profits up to 8000, particularly for clothing types like Dresses, Jackets, and Sweaters.
- At lower quantities (1-3), profits are more scattered, ranging from 0 to 4000, indicating variability in profit margins across clothing types.
- The spread of points suggests that while quantity impacts profit, other factors (e.g., clothing type, pricing) also play a significant role in determining profit.

Justification for Plot Choice:

- A scatter plot is a good choice for this bivariate analysis because it visualizes the relationship between two numerical variables (Quantity and Sales Profit) while incorporating a categorical variable (Clothing Type) through color.
- It helps identify trends and variations, which are important for understanding how quantity influences profit across different products.

Figure 3



Analysis:

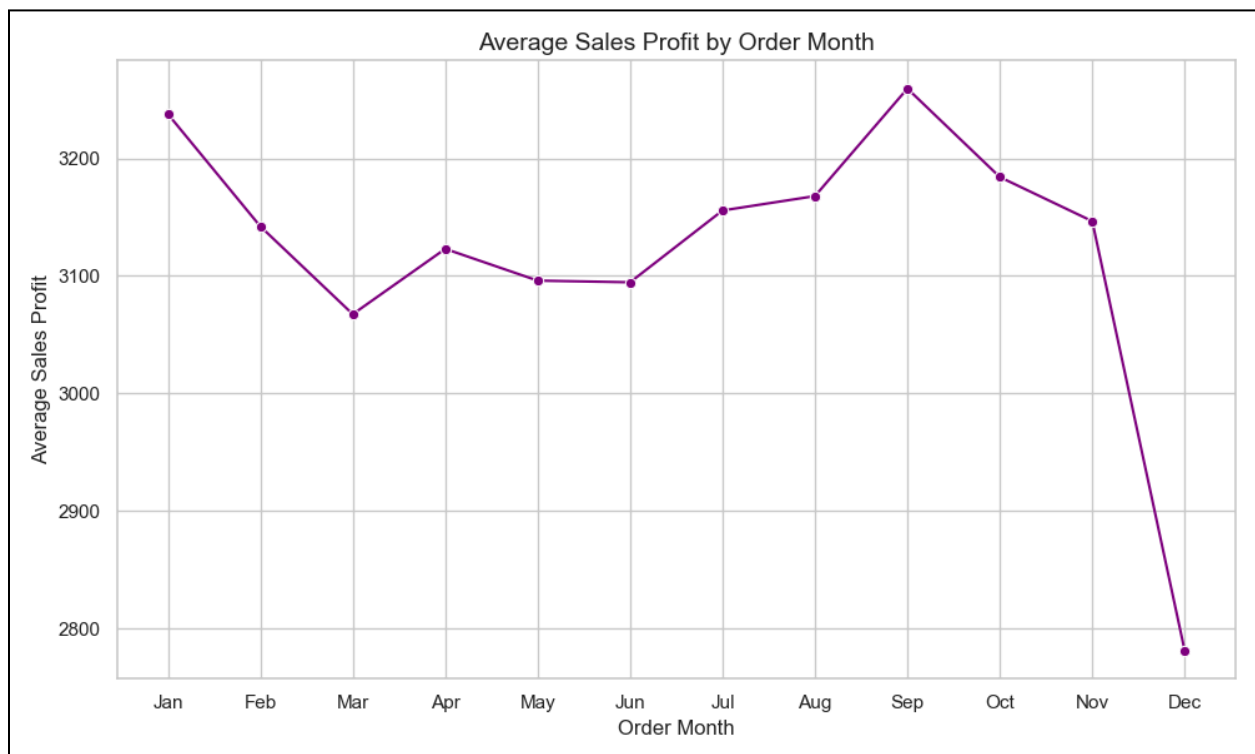
- This boxplot displays the distribution of Sales Profit for each clothing type.
- Sweaters and Shirts have the highest median profits, around 3000, indicating they are more profitable per sale compared to other types.
- Hoodies, shoes and T-shirts show outliers, suggesting occasional high-profit sales despite lower medians.

- Dresses, Skirts, Jackets, Tops, Shoes, Jeans, and Pants have median profits around 2500, with less variation compared to Sweaters and Shirts.
- The presence of outliers in Hoodies, shoes and T-shirts indicates potential for high-profit sales, which could be leveraged through targeted marketing or promotions.

Justification for Plot Choice:

- A boxplot is a good choice for this univariate analysis because it summarizes the distribution of Sales Profit for each clothing type, showing the median, interquartile range, and outliers. This helps identify which clothing types are consistently profitable and which have potential for high-profit sales, aiding in inventory and pricing decisions.

Figure 4



Analysis:

- This line plot shows the average sales profit across each month of the year.

- January starts with the highest average profit (around 3250), followed by a decline in February and March (around 3100).
- Profits stabilize between April and June (around 3100), then peak again in July (around 3200) and August (around 3150).
- A sharp decline occurs from September to December, with December showing the lowest average profit (around 2800) and September showing the highest(around 3280).
- The trend suggests seasonal variations, with higher profits at the beginning of the year and mid-year, possibly due to seasonal sales or festivals, and a drop toward the end of the year, which may require further investigation into customer behavior or external factors.

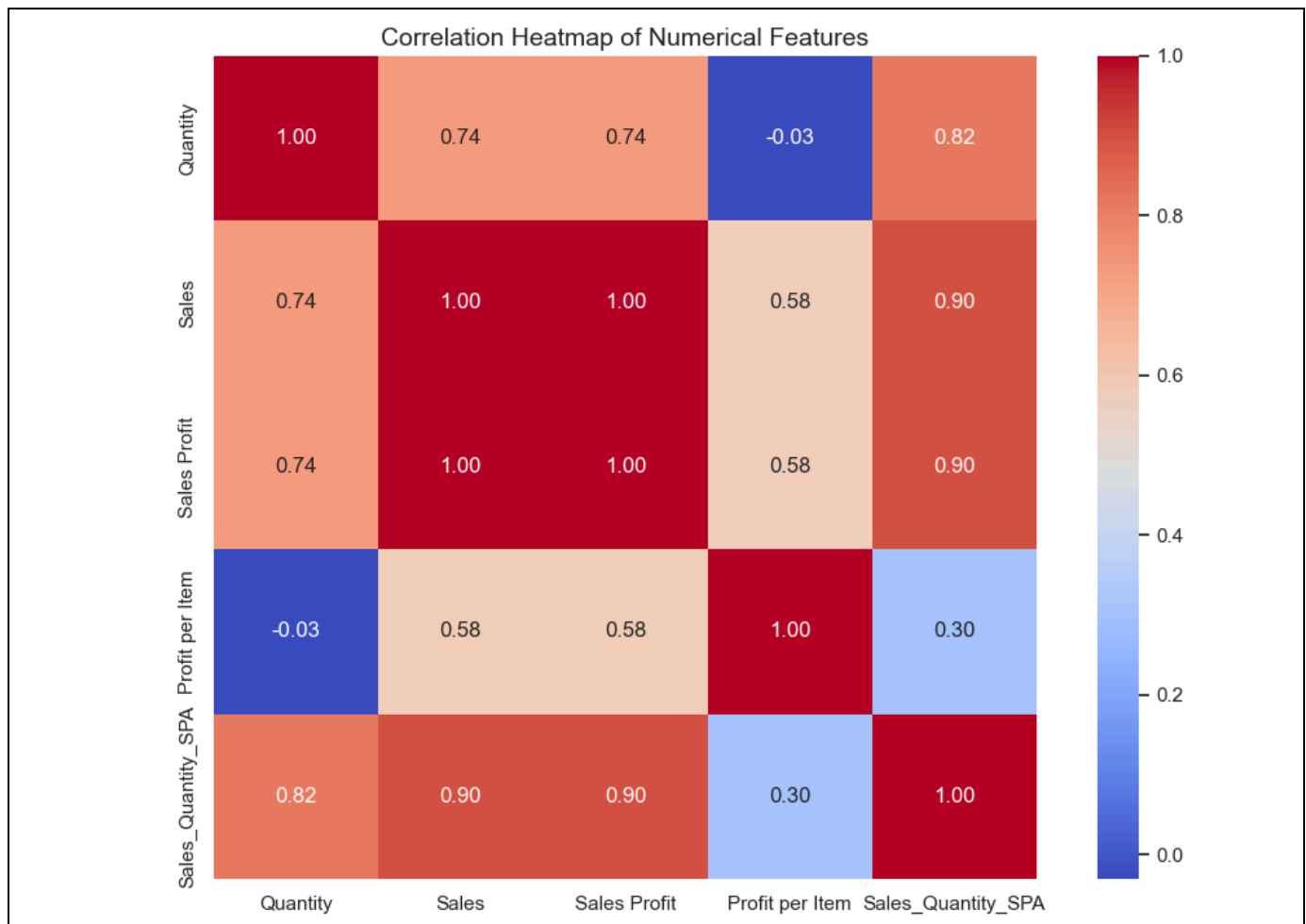
Justification for Plot Choice:

- A line plot is suitable for this bivariate analysis because it effectively shows the trend of a numerical variable (Average Sales Profit) over a categorical variable (Order Month). It highlights seasonal patterns and fluctuations, which are important for understanding sales performance and planning inventory or promotions.

4. Multivariate analysis

- Multivariate analysis examines three or more variables simultaneously to understand complex relationships, often using groupings or interactions.

Figure 1



Analysis:

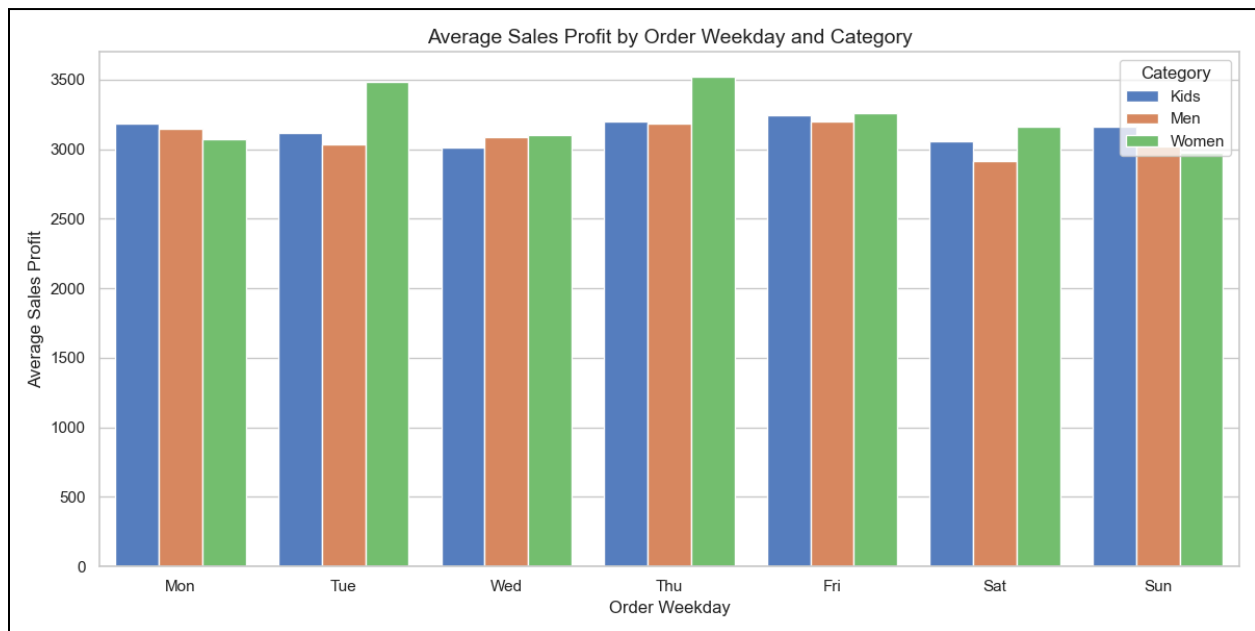
- This heatmap visualizes the correlation between numerical features: Quantity, Sales, Sales Profit, Profit per Item, and Sales_Quantity_SPA.
- Strong positive correlations (1.00) are observed between Sales and Sales Profit, indicating that as sales increase, profit increases proportionally. This is expected in retail data where profit is often a function of sales.

- Quantity shows a strong positive correlation with both Sales (0.74) and Sales Profit (0.74), suggesting that higher quantities sold lead to higher sales and profits.
- Sales_Quantity_SPA has a very strong correlation with Quantity (0.82), Sales (0.90), and Sales Profit (0.90), confirming its relevance as a derived feature for predicting profit.
- Profit per Item has a weak negative correlation with Quantity (-0.03), indicating that selling more items doesn't necessarily increase profit per item, possibly due to bulk discounts or lower margins.
- Profit per Item also shows a moderate positive correlation with Sales (0.58) and Sales Profit (0.58), suggesting some relationship but not as strong as other features.

Justification for Plot Choice:

- A heatmap is an excellent choice for multivariate analysis of numerical features because it provides a quick, color-coded visual representation of correlations.
- It allows for easy identification of strong and weak relationships between multiple variables, which is crucial for feature selection and understanding interdependencies in the dataset.

Figure 2



Analysis:

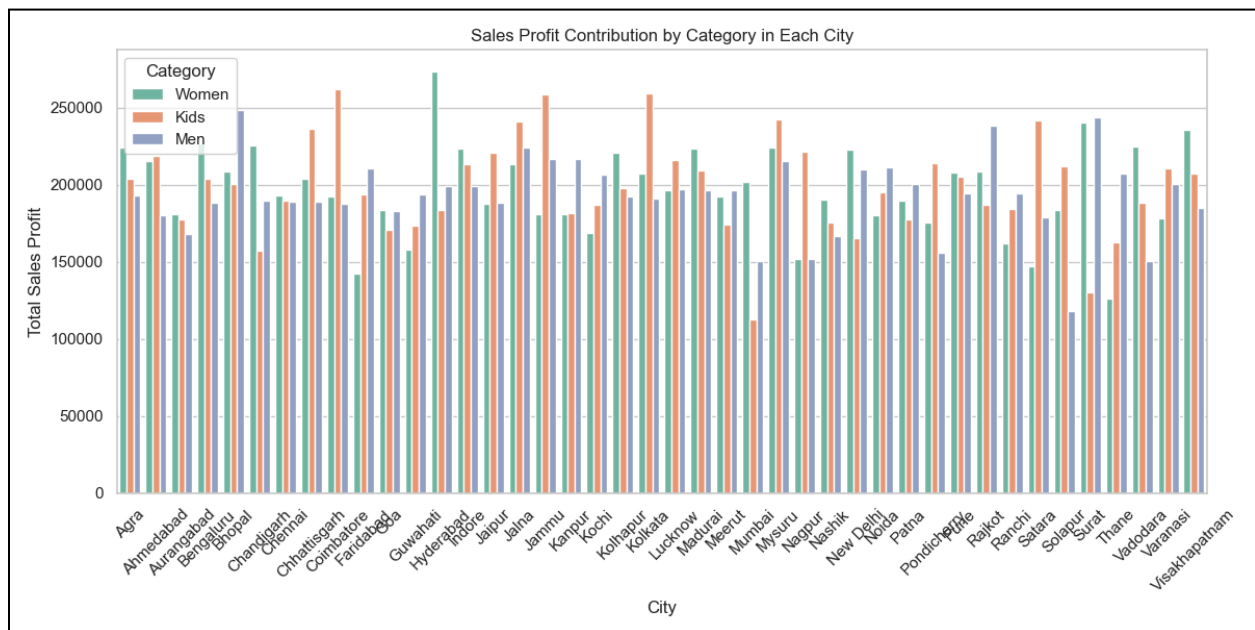
- This grouped bar chart shows the average sales profit across weekdays (Mon to Sun) for three categories: Kids, Men, and Women.
- Kids (blue bars) consistently show higher profits across all weekdays, peaking on Thursday and Friday (around 3500).
- Women (green bars) generally have the second-highest profits, with a notable peak on Tuesday and Thursday (around 3500) and a dip on Sunday (around 3100).
- Men (orange bars) have values hovering around 3100 to 3200, showing less variation compared to the other categories.
- The chart highlights that weekends (Saturday and Sunday) do not necessarily yield the highest profits, with midweek days (Wednesday and Thursday) showing higher profits for all categories.

- This suggests that consumer purchasing behavior may be influenced by midweek shopping trends, possibly due to promotions or specific shopping habits for kids' clothing.

Justification for Plot Choice:

- A grouped bar chart is ideal for this multivariate analysis because it allows comparison across two categorical variables (Order Weekday and Category) with respect to a numerical variable (Average Sales Profit).
- The grouping by weekday and the color-coding by category make it easy to identify patterns and differences, which is useful for understanding sales trends and planning marketing strategies.

Figure 3



Analysis:

- This grouped bar chart shows the total sales profit contribution by category (Women, Kids, Men) across different cities.
- In Ahmedabad, Kids (orange) lead with the highest profit contribution, followed by women and Men.

- Coimbatore shows Kids (orange) contributing the most profit, indicating a strong demand for kids' clothing in this city.
- Cities like Kolhapur and Jalna show balanced contributions across all categories, with profits around 20,000 each.
- Some cities, like Solapur, have lower overall contributions, with Men (blue) showing the least profit in most regions.
- The chart highlights regional preferences, with Women dominating in some cities and Kids in others, which can guide targeted marketing and inventory decisions.

Justification for Plot Choice:

- A grouped bar chart is ideal for this multivariate analysis because it compares a numerical variable (Total Sales Profit) across two categorical variables (Category and City).
- The grouping by city and color-coding by category make it easy to identify regional preferences and category performance, which is crucial for strategic planning in retail.

Methodology

1. Machine Learning Models

- Models Used: The project implements two regression models
 - Linear Regression is a simple and widely used machine learning algorithm that models the relationship between independent variables and a continuous target variable using a straight line. It assumes a linear correlation between features and the target and minimizes the residual error between predicted and actual values by fitting the best line through the data. This model is easy to interpret and serves as a strong baseline for comparison. However, it struggles to capture complex or non-linear relationships, which are often present in real-world retail data.
 - RandomForestRegressor is an ensemble learning algorithm that constructs multiple decision trees and combines their predictions to improve accuracy and reduce overfitting. Each tree is trained on a random subset of the data, and the final output is typically the average of all tree predictions. This model is well-suited for handling non-linear relationships and noisy data, making it robust across various datasets. However, it can be computationally intensive, especially with large feature sets or deep trees, and may require tuning to prevent overfitting.
 - XGBoost (Extreme Gradient Boosting) is a high-performance gradient boosting framework that builds trees sequentially, with each tree aiming to correct the errors made by the previous ones. It optimizes a regularized loss function, which helps reduce overfitting and improve generalization. XGBoost is known for its speed, scalability, and ability to handle missing data internally. It performs exceptionally well on structured datasets but can be sensitive to hyperparameter settings,

making it prone to overfitting if not properly tuned.

- LightGBM is a gradient boosting framework designed for efficiency and scalability. It uses histogram-based algorithms and grows trees leaf-wise rather than level-wise, which allows it to reduce loss more effectively. LightGBM is particularly suitable for large datasets due to its fast training speed and low memory consumption. It can handle categorical features directly and is highly optimized for performance. However, it may overfit on smaller datasets and demands careful hyperparameter tuning to achieve optimal results.
- CatBoost is a gradient boosting algorithm specifically optimized for handling categorical features. It employs ordered boosting and uses advanced techniques to encode categorical variables internally without needing explicit preprocessing, thus reducing the risk of data leakage. CatBoost is highly accurate and performs well with default parameters, making it a strong choice for datasets with many categorical variables. Its primary drawback is slower training speed on purely numerical datasets compared to LightGBM.
- Justification:
 - Linear Regression: Provides a simple baseline for comparison..
 - Ensemble Methods(Random Forest,XG Boost,Light GBM,Cat Boost): Chosen for their ability to model non-linear relationships and handle retail data complexities like categorical variables and interactions
 - Diversity: Covers a range of approaches (linear, tree-based, boosting) to ensure comprehensive evaluation.

2. Model Implementation

- Training and Testing: The dataset was split into 80%(6296 rows) for training and 20%(1574 rows) for testing using train_test_split(from

scikit-learn with random_state=42 for reproducibility).

- Cross-Validation: 5-fold cross-validation was used. Dataset split into 5 folds; each fold used as a validation set while training on the remaining 4. R^2 score computed for each fold, then averaged.
- Hyperparameter tuning : carried out using GridSearchCV from the scikit-learn library. This method performs an exhaustive search over specified parameter values and evaluates each combination using cross-validation. It automates the tuning process and ensures consistent evaluation through systematic experimentation.

Model	Parameter	Values tested	Description
Random Forest	n_estimators	50, 100, 200	Number of trees in the forest
	max_depth	10, 20, None	Maximum depth of each tree
	min_samples_split	2,5	Minimum samples required to split an internal node
XGBoost	n_estimators	100, 200	Number of boosting rounds (trees)
	learning_rate	0.01, 0.1	Shrinks feature contributions in each boosting step
	max_depth	3,6,9	Maximum tree depth for base

			learners
LightGBM	n_estimators	100,200	Number of boosting rounds
	learning_rate	0.01, 0.1	Controls step size in updates
	num_leaves	31, 50	Maximum number of leaves in one tree
CatBoost	iterations	200, 500	Number of boosting iterations
	learning_rate	0.01, 0.1	Controls the speed of learning
	depth	4,6,8	Depth of the tree

3. Feature Selection and Extraction

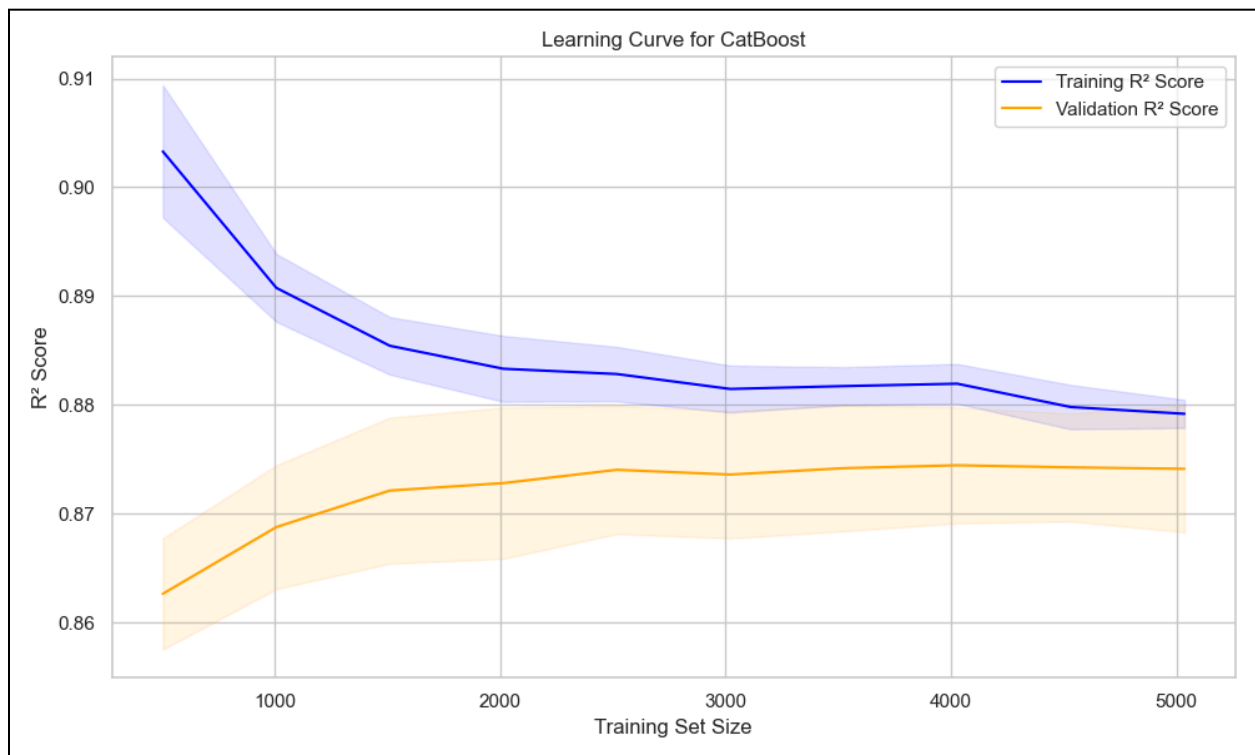
- Method Used: RandomForest Feature Importance=> trained a Random Forest model, extracted feature importance scores.
- Sorted the features by importance and selected only those contributing to nearly 80% of cumulative importance. Focus was on most impactful features to reduce model complexity and overfitting risk.
- Sales_Quantity_SPA (importance score: 0.32) was selected as the important feature.

4. Model Validation

Learning curves plot a model's performance (e.g., R^2 score) on the training and validation sets as the training set size increases. They help diagnose:

- Underfitting: Occurs when both training and validation scores are low and don't improve much with more data, indicating the model is too simple to capture the data's patterns.
- Overfitting: Occurs when the training score is high but the validation score is low, with a large gap between the two, indicating the model is too complex and memorizes the training data rather than generalizing to unseen data.
- Good Fit: Occurs when the training and validation scores are both high and converge as the training set size increases, indicating the model generalizes well.

Figure 1



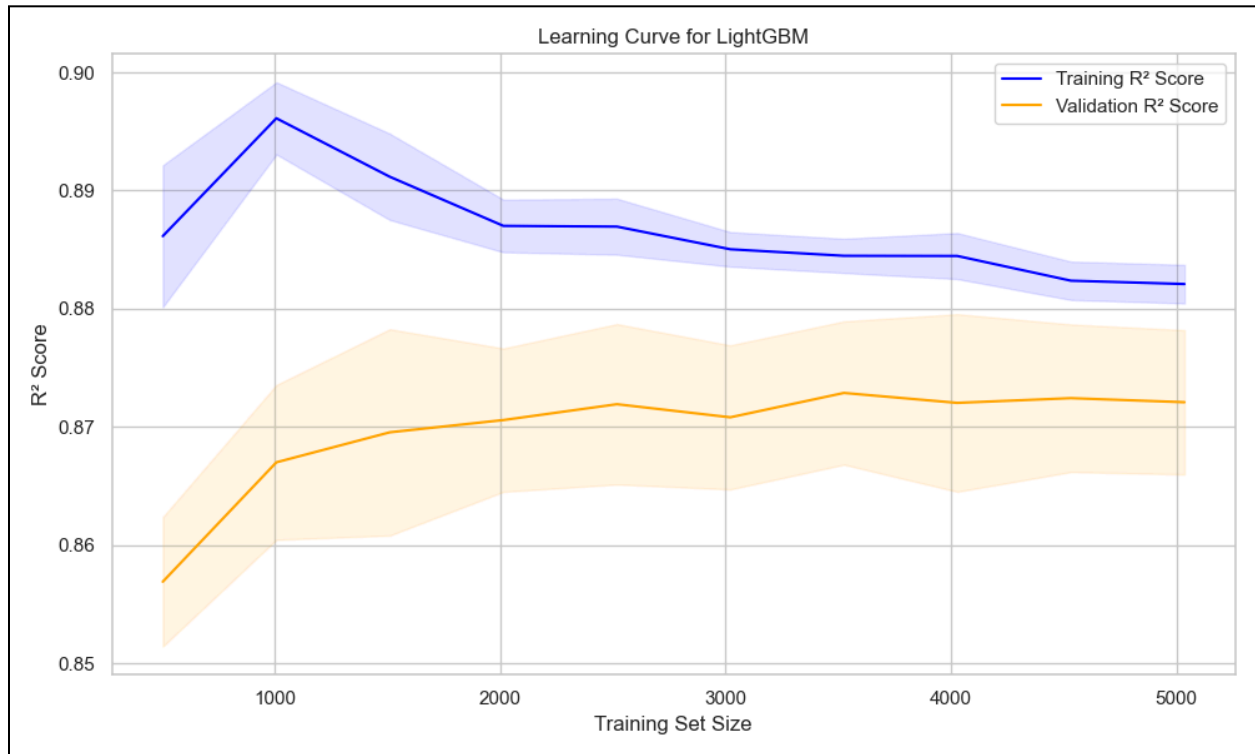
Analysis:

- The training R^2 score starts high (around 0.91) with a small training set (1000 samples) but decreases gradually to around 0.88 as the training set size increases to 5000.
- The validation R^2 score starts at 0.87, increases slightly, and stabilizes around 0.88, converging with the training score.
- The shaded areas represent the variance in scores across different folds (likely from cross-validation), showing that both training and validation scores have low variance, indicating stability.
- The curves converge at an R^2 score of approximately 0.88, suggesting that CatBoost generalizes well to unseen data.

Underfitting/Overfitting Diagnosis:

- No Underfitting: Both training and validation scores are relatively high (around 0.88), indicating that the model is not too simple and captures the data's patterns effectively.
- Minimal Overfitting: Initially, there is a small gap between the training and validation scores (0.91 vs. 0.87), suggesting slight overfitting with smaller datasets. However, as the training set size increases, the gap narrows, and the scores converge, indicating that overfitting is mitigated with more data.
- Good Fit: The convergence of the training and validation scores at a high R^2 value (0.88) shows that CatBoost achieves a good balance between fitting the training data and generalizing to the validation

Figure 2



Analysis:

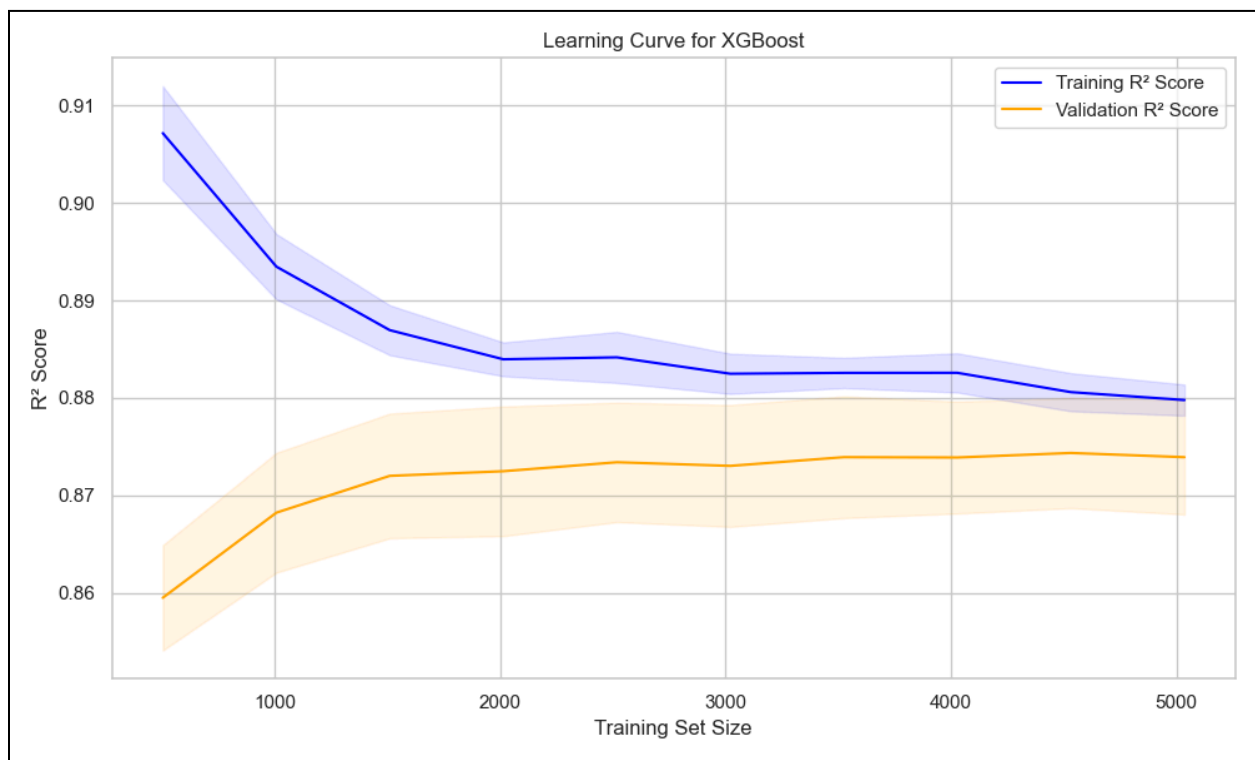
- The training R^2 score starts at 0.90 with 1000 samples, decreases to around 0.88, and stabilizes as the training set size increases to 5000.
- The validation R^2 score starts at 0.87, increases slightly, and stabilizes around 0.88, closely following the training score.
- The variance (shaded areas) is low for both training and validation scores, indicating stable performance across folds.
- The curves converge at an R^2 score of approximately 0.88, similar to CatBoost, showing good generalization.

Underfitting/Overfitting Diagnosis:

- No Underfitting: Both training and validation scores are high (around 0.88), indicating that LightGBM captures the data's patterns well and is not too simple.

- Minimal Overfitting: There is a small initial gap between the training and validation scores (0.90 vs. 0.87), suggesting slight overfitting with smaller datasets. However, the gap narrows as the training set size increases, and the scores converge, showing that overfitting is reduced with more data.
- Good Fit: The convergence of the training and validation scores at a high R^2 value (0.88) indicates that LightGBM generalizes well, similar to CatBoost.

Figure 3



Analysis:

- The training R^2 score starts at 0.91 with 1000 samples, decreases steadily to around 0.88, and stabilizes as the training set size increases to 5000.
- The validation R^2 score starts at 0.87, increases slightly, and stabilizes around 0.88, converging with the training score.
- The variance in scores is low, indicating stable performance across folds.
- The curves converge at an R^2 score of approximately 0.88, consistent with CatBoost and LightGBM, showing good generalization.

Underfitting/Overfitting Diagnosis:

- No Underfitting: Both training and validation scores are high (around 0.88), indicating that XGBoost is not too simple and captures the data's patterns effectively.
- Minimal Overfitting: A small initial gap exists between the training and validation scores (0.91 vs. 0.87), suggesting slight overfitting with smaller datasets. The gap narrows as the training set size increases, and the scores converge, indicating that overfitting is mitigated with more data.
- Good Fit: The convergence at a high R^2 score (0.88) shows that XGBoost achieves a good balance between fitting the training data and generalizing to the validation set.

Figure 4



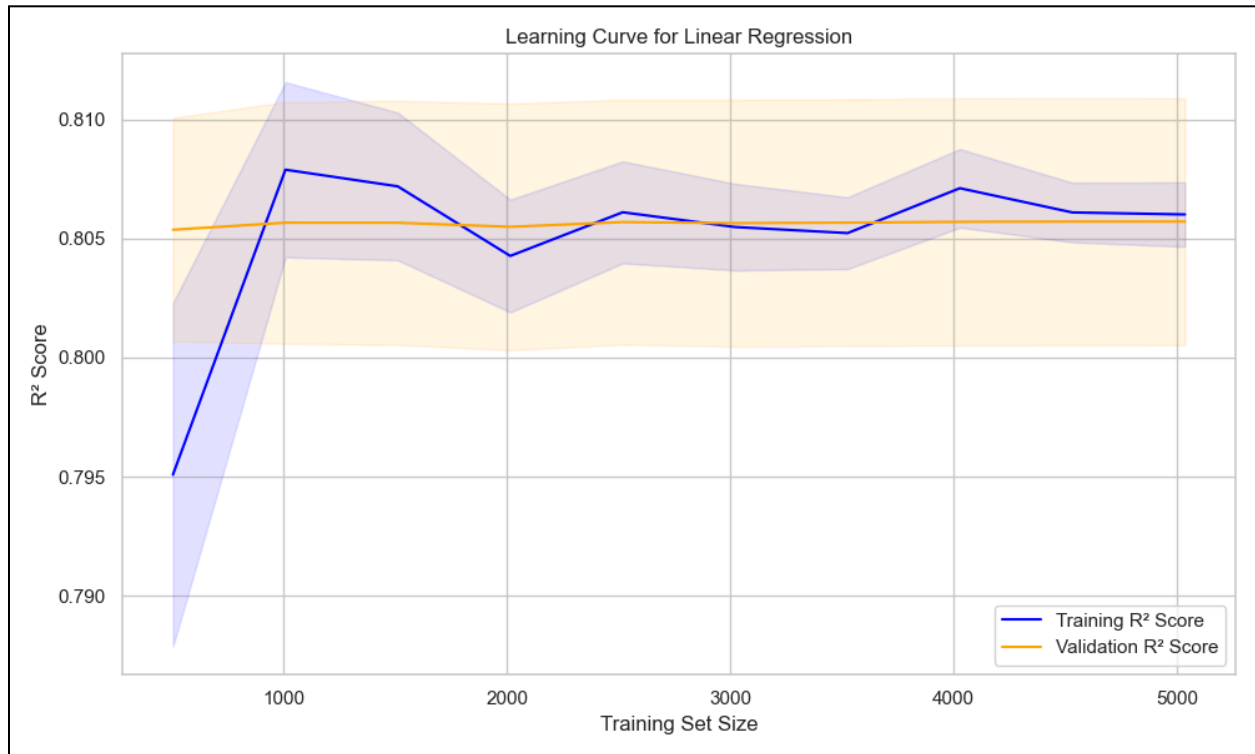
Analysis:

- The training R^2 score starts very high (around 0.96) with 1000 samples, decreases steadily to around 0.92 as the training set size increases to 5000, but remains significantly higher than the validation score.
- The validation R^2 score starts at 0.84, increases slightly, and stabilizes around 0.86, showing a consistent gap with the training score.
- The variance in scores is low, indicating stable performance, but the gap between the training and validation scores persists.
- The curves do not fully converge, with the training score remaining higher (0.92) than the validation score (0.86), suggesting some overfitting.

Underfitting/Overfitting Diagnosis:

- No Underfitting: The validation score is reasonably high (0.86), indicating that Random Forest captures the data's patterns and is not too simple.
- Overfitting Present: A noticeable gap exists between the training and validation scores (0.92 vs. 0.86), even as the training set size increases. This gap indicates that Random Forest is overfitting, as it performs much better on the training data than on the validation data, suggesting it may be memorizing the training data rather than generalizing well.
- Fit Assessment: While the validation score is decent, the persistent gap suggests that Random Forest is not generalizing as well as the boosting models (CatBoost, LightGBM, XGBoost).

Figure 5



Analysis:

- The training R^2 score starts at 0.79 with 1000 samples, increases to around 0.81, and stabilizes around 0.805 as the training set size increases to 5000.
- The validation R^2 score starts at 0.79, fluctuates slightly, and stabilizes around 0.805, closely following the training score.
- The variance in scores is low, indicating stable performance across folds.
- The curves converge at an R^2 score of approximately 0.805, but this value is lower than that of the other models, suggesting limited predictive power.

Underfitting/Overfitting Diagnosis:

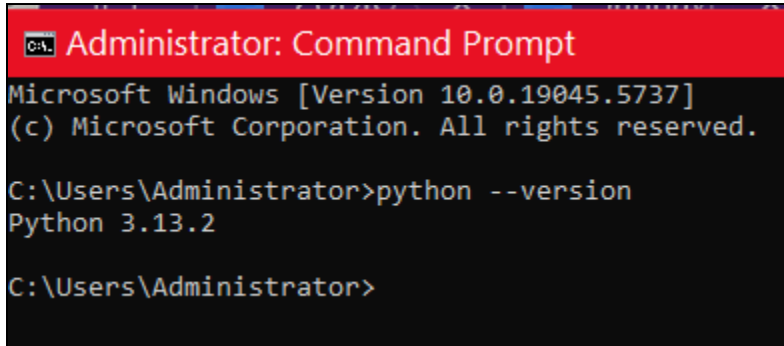
- Underfitting Present: Both the training and validation scores are relatively low (around 0.805) compared to the other models (0.88). This indicates that Linear Regression is too simple to capture the complex, non-linear patterns in the retail sales data, leading to underfitting.

- No Overfitting: The training and validation scores are very close and converge quickly, with no significant gap, indicating that Linear Regression does not overfit. However, this comes at the cost of poor performance.
- Fit Assessment: The low R^2 score (0.805) confirms that Linear Regression is not a good fit for this dataset, as it fails to capture the underlying patterns, consistent with the report's findings of heteroscedasticity in residuals (indicating non-linear relationships that Linear Regression cannot model).

Experimental Setup

Software used :

- Version: python 3.13.2 has been used to do this project.

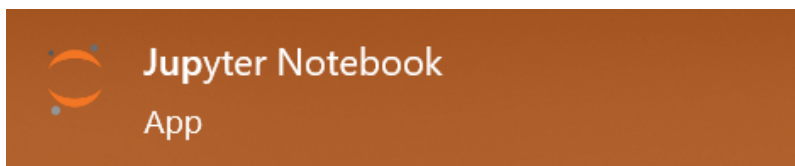


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Administrator: Command Prompt
Microsoft Windows [Version 10.0.19045.5737]
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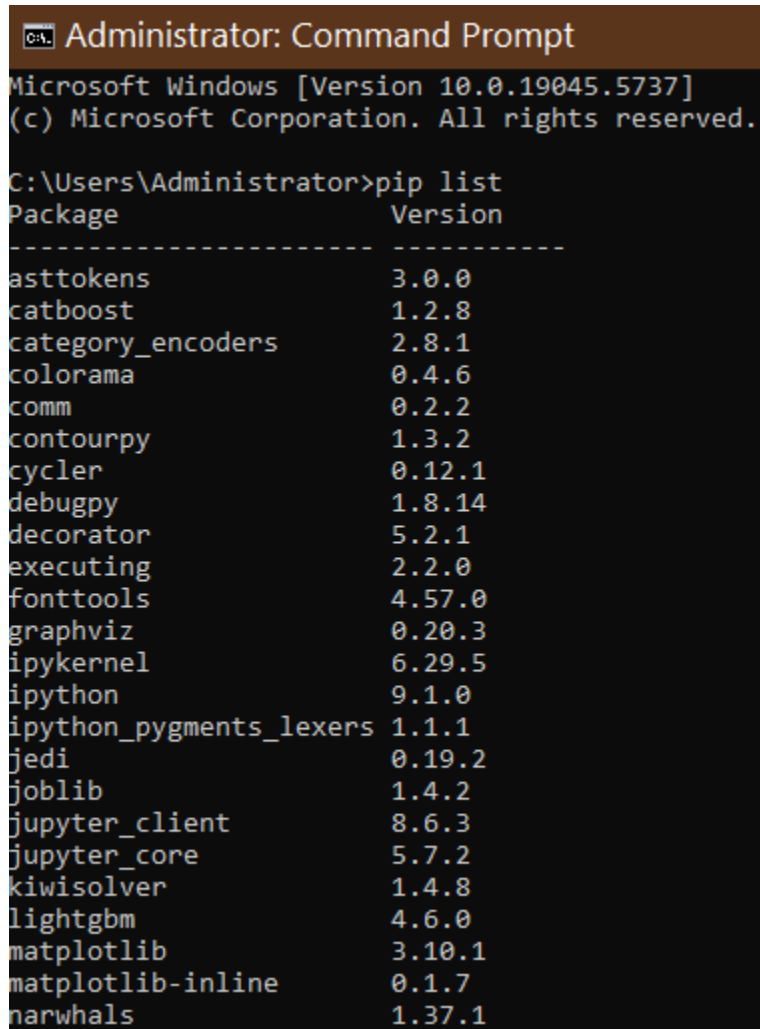
C:\Users\Administrator>python --version
Python 3.13.2

C:\Users\Administrator>
```

- IDE: Pycharm and jupyter notebook



- Libraries:



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C:\Users\Administrator>pip list
Package                Version
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asttokens               3.0.0
catboost                1.2.8
category_encoders       2.8.1
colorama                0.4.6
comm                    0.2.2
contourpy               1.3.2
cyclcr                  0.12.1
debugpy                 1.8.14
decorator                5.2.1
executing               2.2.0
fonttools               4.57.0
graphviz                0.20.3
ipykernel                6.29.5
ipython                 9.1.0
ipython_pygments_lexers 1.1.1
jedi                    0.19.2
joblib                  1.4.2
jupyter_client          8.6.3
jupyter_core            5.7.2
kiwisolver              1.4.8
lightgbm                4.6.0
matplotlib              3.10.1
matplotlib-inline       0.1.7
narwhals                 1.37.1
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nest-asyncio      1.6.0
numpy             2.1.3
packaging         25.0
pandas            2.2.3
parso             0.8.4
patsy             1.0.1
pillow            11.2.1
pip               24.3.1
platformdirs     4.3.7
plotly            6.0.1
prompt_toolkit    3.0.51
psutil            7.0.0
pure_eval         0.2.3
Pygments          2.19.1
pyparsing         3.2.3
python-dateutil   2.9.0.post0
pytz              2025.1
pywin32           310
pyzmq             26.4.0
scikit-learn      1.6.1
scipy             1.15.2
six               1.17.0
stack-data        0.6.3
statsmodels       0.14.4
threadpoolctl     3.6.0
tornado           6.4.2
traitlets         5.14.3
tzdata            2025.1
wcwidth           0.2.13

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xgboost           3.0.0

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- pandas as pd: For data manipulation and analysis using DataFrames.
- numpy as np: For numerical operations, especially arrays and mathematical functions.
- matplotlib.pyplot as plt: For creating static visualizations like line plots and bar charts.
- seaborn as sns: For advanced and attractive statistical data visualizations built on matplotlib.
- train_test_split, cross_val_score: To split data into training/testing sets and evaluate models using cross-validation.
- OneHotEncoder, StandardScaler: For encoding categorical variables and scaling numerical features.

- ColumnTransformer: Applies different preprocessing to selected columns in a pipeline.
- RandomForestRegressor: An ensemble model using multiple decision trees for regression tasks.
- LinearRegression: A linear model to predict continuous target variables.
- r2_score: Evaluation metric to measure how well predictions match the actual values (coefficient of determination).
- warnings: Used to suppress warnings during code execution.
- GridSearchCV: Performs exhaustive search over parameter values for model tuning using cross-validation.
- XGBRegressor: An efficient and accurate gradient boosting regressor from the XGBoost library.
- LGBMRegressor: A fast, memory-efficient gradient boosting model from LightGBM.
- CatBoostRegressor: A gradient boosting model optimized for handling categorical features from CatBoost.
- TargetEncoder: Encodes categorical variables based on the target mean (useful for high-cardinality features).
- learning_curve: Used to visualize how model performance evolves with increasing training data.

Hardware used :

```
C:\Users\Administrator>systeminfo

Host Name:                DESKTOP-DDRKGLH
OS Name:                   Microsoft Windows 10 Pro
OS Version:                10.0.19045 N/A Build 19045
OS Manufacturer:          Microsoft Corporation
OS Configuration:         Standalone Workstation
OS Build Type:              Multiprocessor Free
Registered Owner:          Admin
Registered Organization:
Product ID:                 00330-81477-13618-AA351
Original Install Date:      11/19/2022, 12:20:52 AM
System Boot Time:           4/18/2025, 2:39:50 PM
System Manufacturer:        Hewlett-Packard
System Model:                HP 240 G8 Notebook PC
System Type:                 x64-based PC
Processor(s):                1 Processor(s) Installed.
                             [01]: Intel64 Family 6 Model 140 Stepping 1 GenuineIntel ~2419 Mhz
BIOS Version:                AMI F.07, 10/22/2021
Windows Directory:           C:\Windows
System Directory:            C:\Windows\system32
Boot Device:                 \Device\HarddiskVolume1
System Locale:                en-us;English (United States)
Input Locale:                 00004009
Time Zone:                   (UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi
Total Physical Memory:        7,859 MB
Available Physical Memory:    1,306 MB
Virtual Memory: Max Size:    14,572 MB
Virtual Memory: Available:    3,011 MB
Virtual Memory: In Use:       11,561 MB
Page File Location(s):        C:\pagefile.sys
Domain:                       WORKGROUP
Logon Server:                  \\DESKTOP-DDRKGLH
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Evaluation Metrics :

In this project, we used two key evaluation metrics to assess the performance of our machine learning models: R-squared (R^2) and Cross-Validation R-squared.

1. R-squared (R^2): This metric measures how well the regression model explains the variability of the target variable (Sales Profit). It ranges from 0 to 1, where a higher value indicates a better fit. An R^2 of 1 means perfect prediction, while 0 means the model fails to capture any trend.
2. Cross-Validation R^2 (Average R^2 across 5-folds): To ensure the model generalizes well to unseen data, we applied 5-fold cross-validation using the `cross_val_score` function from Scikit-learn. This method splits the dataset

into 5 parts, trains the model on 4 parts, and tests it on the remaining part—repeating this 5 times. The average of the 5 R^2 scores gives a more reliable performance estimate.

These metrics help compare different models and select the one that provides the best predictive accuracy.

Results and Discussion

Performance Comparison:

- In this project, we implemented 5 machine learning models to predict the sales profit of Zudio retail stores. The models were evaluated using the R^2 Score and 5-fold Cross-Validation R^2 .

ML Model	R^2 score	Avg. cross validation R^2
CatBoost	0.880690	0.874184
XGBoost	0.880781	0.873754
LightGBM	0.879757	0.872109
Random Forest	0.876400	0.868097
Linear Regression	0.804648	0.805719

Best Model: CatBoost

Error Analysis

Underfitting

- Linear Regression: Low R^2 score (0.8057 CV, 0.8046 test) shows underfitting.

Overfitting

- Random Forest: Learning curve shows a gap (training R^2 0.92, validation 0.86), indicating overfitting. Tuning via GridSearchCV (eg. max_depth, min_samples_split) reduced this, but some overfitting persists.

CatBoost (Best Model)

- Performance: Balanced learning curves (training/validation converge at 0.88). CV R^2 of 0.8742 and test R^2 of 0.8807 show good generalization. Residuals show no patterns, and stability check confirms robustness.
- LightGBM/XGBoost: Similar to CatBoost, converge at $R^2 = 0.88$ with low variance, indicating minimal errors (test R^2 : 0.8798, 0.8808).

Conclusion

Summary:

The project focused on analyzing Zudio's retail sales data using a combination of data analytics and machine learning techniques. The process began with extensive data preprocessing, exploratory data analysis (EDA), and advanced feature engineering to enhance data quality and extract meaningful patterns. Multiple regression models were trained, including Linear Regression, Random Forest, XGBoost, LightGBM, and CatBoost. Among these, CatBoost emerged as the best-performing model, achieving an average cross-validation R^2 score of 0.8742, indicating strong predictive capability.

Findings:

- Key Features: The most influential features in driving profit predictions were Sales_Quantity_SPA, Profit per Item, and Quantity, as determined by feature importance analysis.
- Regional Insights: The city of Kolhapur stood out as the top-performing region in terms of sales profit, providing useful input for regional strategy.
- Prediction Example: Using the trained CatBoost model, a hypothetical scenario was predicted to yield a profit of ₹6242.53, demonstrating the model's application for forecasting.
- Applications: The insights derived from the model can support inventory management, targeted marketing, and resource allocation, helping Zudio make data-driven decisions.

Limitations:

While the project has delivered valuable insights, it is important to acknowledge several limitations:

- Data Size: The dataset contained only 7870 rows, which may not fully capture long-term sales patterns or rare events.
- Bias: Certain customer segments or promotional campaigns may not be well-represented, introducing potential bias into the model's predictions.
- Missing Features: The dataset lacked critical variables such as customer demographics, loyalty metrics, or promotional data, limiting the depth of analysis.

Future Work:

The project can be extended and improved in several ways:

- Dataset Expansion: Incorporate a broader dataset including seasonal patterns, economic indicators, and competitor pricing to improve model generalization.
- Advanced Modeling: Explore more complex models like neural networks or recurrent architectures for capturing deeper temporal patterns and interactions.
- Feature Enrichment: Integrate external data sources such as weather, event calendars, and festive seasons to enhance feature engineering.
- Real-Time Forecasting: Develop systems for live prediction using real-time data streams, enabling Zudio to adapt dynamically to changing retail environments.

References

Youtube links:

- ALL ML MODELS EXPLAINED(AI for beginners)
<https://youtu.be/0YdpwSYMY6I?si=EuylikRotHrqpTc9>
- What's machine learning(IBM)
<https://youtu.be/9gGnTQTYNaE?si=FljpXDo6R9MBAdNA>
- Supervised vs unsupervised learning(IBM)
https://youtu.be/W01tIRP_Rqs?si=v8R1dDLbAUyuNjhE
- Python full course for beginners(Coding with Sagar)
<https://youtu.be/gPvYox9JIYI?si=kjF5gjBEbja4R5HZ>
- Data Analyst Bootcamp(Udemy)
https://www.udemy.com/share/103IdG3@vHF4RcEUMQJkVhNys-Scyg4K6m-4xev_FC-zsls-YmlN-2x7O_mgc63DYJmFKLQOkA==/

Websites referred:

- Kaggle
[Kaggle: Your Machine Learning and Data Science Community](#)
- Python documentation
[3.13.3 Documentation](#)

AI used:

- Napkin AI to get the visuals
[Napkin AI - The visual AI for business storytelling](#)
- ChatGPT to enhance the code
[ChatGPT](#)

